

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION

Federal State Autonomous Educational
Institution of Higher Education "Southern Federal University"
(SOUTHERN FEDERAL UNIVERSITY)

COURSE SYLLABUS FOR
Discrete Mathematics (Дискретная математика)

Подписано электронной подписью:

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I. GOALS AND OBJECTIVES OF MASTERING THE DISCIPLINE

Objectives of mastering the discipline:

- formation of a system of knowledge about the methodology of using mathematical tools of discrete mathematics;
- teaching students to build models, methods and algorithms based on set theory, mathematical logic, graph theory and combinatorics; study of the basics of discrete mathematical methods;
- consideration of practical, theoretical and fundamental problems for existence, effective construction, enumeration and optimization of objects that depend on a large number of discrete variables.
- mastering the modern apparatus and methods of discrete mathematics for further use in solving theoretical and applied problems;
- formation of the student's personality, development of his intellect and ability to logical and algorithmic thinking;
- development of the ability to analyze the results obtained, skills of independent study of literature on mathematics and its applications.

Objectives of mastering the discipline:

- study of the basics of discrete mathematics and mastering the methods of solving practical problems on the topics of the discipline;
- development of the ability to operate with the concepts and methods of discipline used in further educational and professional activities;
- to develop students' ability to independently expand and deepen mathematical knowledge, to teach them to use mathematical symbols;
- to demonstrate mathematical models of the simplest systems and processes in natural science and technology, to instill skills in the use of basic methods of processing experimental data;
- to form the skills of working with graphic constructions in mathematics;
- to demonstrate to students the essence of the scientific approach, the specifics of mathematics and its role in the development of science using examples of mathematical concepts and methods of operation of the laws of science.

II. THE PLACE OF THE DISCIPLINE IN THE STRUCTURE OF THE EDUCATIONAL PROGRAM

The discipline belongs to the module of university-wide disciplines of the educational program.

This discipline is based on the basic knowledge, skills and abilities formed when obtaining the previous level of education.

Knowledge, skills and abilities formed by this discipline will be required when mastering the following elements of the educational program:

- higher mathematics, for industrial, pre-diploma practice and final qualification work.

III. REQUIREMENTS FOR THE RESULTS OF MASTERING THE DISCIPLINE

The development of the discipline is aimed at the formation of the following competencies in accordance with the educational standard and educational program:

List of planned learning outcomes in the discipline correlated with indicators of competence achievement

Competence	Indicators of Competency Achievement	Learning outcomes
CPC-1. Is able to understand the natural science essence of problems that arise in the course of professional activity, and apply interdisciplinary and general scientific knowledge and methods to solve the problems of analysis, modeling and research of objects of professional activity	CPC-1.1. Analyzes the natural science essence of problems arising in the course of professional activity	Knowledge: – theoretical foundations of set theory, mathematical logic, graph theory and combinatorics for understanding the natural science essence of problems arising in the course of professional activity
		Skills: – apply the methods of basic tools of set theory, mathematical logic, graph theory, Boolean algebra, elements of combinatorics to analyze natural science problems that arise in the course of professional activity
		Skills: – analysis of natural science problems arising in the course of professional activity using the basic tools of set theory, mathematical logic, graph theory and combinatorics
	CPC-1.2. Solves standard professional problems using interdisciplinary and general scientific knowledge, methods of mathematical analysis and modeling	Knowledge: – the basics of the theory of logic and sets, graph theory, Boolean algebra, elements of combinatorial analysis for understanding standard professional problems in the context of interdisciplinary and general scientific knowledge
		Skills: – apply combinatorial configurations to solve problems, determine the type of binary relations and their properties, perform operations on sets, perform operations on graphs, perform identical transformations, find SDNF, SCNF, determine the minimum DNF for solving both standard professional problems and interdisciplinary problems
		Skills: – possession of the basic tools of discrete mathematics, mathematical logic for solving standard professional problems using interdisciplinary and general scientific knowledge

Competence	Indicators of Competency Achievement	Learning outcomes
PC-5. MF-5 Capable of applying advanced mathematical techniques (graph theory, functional analysis, category theory) to solve complex AI problems and develop new algorithms.	PC-5.1 MF-5.1 Applies modern theoretical mathematics to develop new algorithms and formulate promising AI problems	Knowledge: - basic concepts of graph theory; - basic algorithms of graph theory, including algorithms for traversing graphs, finding the shortest paths, coloring graphs, etc.; - basic characteristics for the spectral analysis of the graph; - about the concept of a random graph.
		Skills: - apply the basic concepts, characteristics and algorithms of graph theory to solve complex AI problems, including the analysis of the structure of neural networks
		Skills: - Mastery of the basic tools of graph theory for solving complex AI problems and developing new algorithms

IV. CONTENT AND STRUCTURE OF THE DISCIPLINE

The complexity of the discipline is 6 credits, 216 hours,
including 6 credits, 216 hours
1 credit, 36 hours per exam

Form of intermediate certification: exam

4.1. The content of the discipline structured by topics (study meetings) for _1_ semester of full-time study

No p/n	Topic of the lesson and issues under consideration	Semester	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
			Contact work				Independen t work				
			Lectures	Practical exercises	Laboratory Classes	Consulting					
Module 1. Fundamentals of Set Theory											
1	The Concept of Set	1	2	–	–	-	1	Informational lecture	–		–
2	Set Setting	1	-	2	–	-	1	Seminar	–		–
3	Boolean Operations	1	2	-	–	-	1	Informational lecture	-	-	-
4	Boolean functions	1	-	2	–	-	1	Seminar	–	-	–
5	Set operations	1	2	–	–	-	1	Informational lecture	-	-	–
6	Methods of mutual inclusion and from the opposite	1	–	2	–	-	1	Seminar	-	-	-
7	Direct product of sets	1	2	–	–	-	1	Informational lecture	–	-	–
8	Sets, equality of sets	1	–	2	–	-	3	Seminar	Practical Work No1	Practical work	6
9	Inversion and composition	1	2	-	-	-	1	Informational lecture	-	-	-
10	Equality of Direct Products	1	-	2	-	-	1	Seminar	-	-	-

No p/n	Topic of the lesson and issues under consideration	Semester	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
			Contact work				Independen t work				
			Lectures	Practical exercises	Laboratory Classes	Consulting					
11	Correspondences. Mapping operations	1	2	-	-	-	1	Informational lecture	-	-	-
12	The image and prototype of the multitude	1	-	2	-	-	2	Seminar	-	-	-
13	Mapping properties	1	2	-	-	-	1	Informational lecture	-	-	-
14	Features	1	-	2	-	-	1	Seminar	-	-	-
15	Relationships. Relationship Operations	1	2	-	-	-	1	Informational lecture	-	-	-
16	Relationship Equality	1	-	2	-	-	1	Seminar			
17	Basic Properties of Relationships	1	2	-	-	-	1	Informational lecture	-	-	-
18	Basic Properties of Relationships	1	-	2	-	-	1	Seminar	-	-	-
19	Fuzzy sets. Operations on fuzzy sets	1	2	-	-	-	1	Informational lecture	-	-	-
20	Inclusion and equality of fuzzy sets	1	-	2	-	-	2	Seminar	-	-	-
21	Fuzzy sets. Operations on fuzzy sets	1	2	-	-	-	1	Informational lecture	-	-	-
22	Boundary control on the basics of set theory		-	2	-	-	4	Seminar	Test No1	Test	14
Module 2. Algebra of Logic											
23	Elements of the Algebra of Logic. Boolean functions	1	2	—	—	-	1	Informational lecture	—	-	—
24	Work with basic logical identities and laws	1	-	2	—	-	1	Seminar	—	-	—
25	Normal Forms of Boolean Functions	1	2	-	—	-	1	Informational lecture	-	-	-

No p/n	Topic of the lesson and issues under consideration	Semester	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
			Contact work				Independen t work				
			Lectures	Practical exercises	Laboratory Classes	Consulting					
26	K-variable decomposition, SDNF, SCNF	1	-	2	—	-	1	Seminar	—	-	—
27	Minimization of Boolean functions. Logic Circuits	1	2	—	—	-	1	Informational lecture	—	-	—
28	Construction of Carnot maps of 2, 3, 4 variables. Logic Circuits	1	—	2	—	-	1	Seminar	-	-	-
29	Minimizing Boolean Functions	1	2	—	—	-	1	Informational lecture	—		—
30	Methods of Uncertain Odds, Quine McClasskey	1	-	2	-	-	3	Seminar	Practical Work No2	Practical work	6
31	Minimization of polynomial SNF and in the Webb basis. The concept of a formal system.	1	2	-	-	-	1	Informational lecture	-	-	-
32	Minimization methods in polynomial SNF and in Webb's basis.	1	-	2	-	-	1	Seminar	-	-	-
33	Post's theorem on functional completeness. 5 classes of Boolean functions.	1	2	-	-	-	1	Informational lecture	-	-	-
34	Boundary Control in Set Theory and Algebra of Logic	1	—	2	—	-	4	Seminar	Test No2	Test	14
Module 3. Fundamentals of graph theory and combinatorics											
35	Basic concepts of graph theory and their interpretation. Fuzzy graph.	1	2	—	—	-	1	Informational lecture	—	-	—
36	Types of graphs. Routes, chains, cycles, bridges, and hinges. Fuzzy graph.	1	—	2	—	-	1	Seminar	-	-	-

No p/n	Topic of the lesson and issues under consideration	Semester	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
			Contact work				Independen t work				
			Lectures	Practical exercises	Laboratory Classes	Consulting					
37	Circumvention of the count. Special graph cycles (Euler and Hamiltonian). Graph cliques, isomorphism	1	2	–	–	-	1	Informational lecture	-	-	-
38	Circumvention of the count. Construction of Euler and Hamiltonian chains and cycles. Graph cliques, isomorphism	1	–	2	–	-	1	Seminar	-	-	-
39	Distance on the graph. Algorithms for finding the shortest paths. Graph metrics.	1	2	–	–	-	1	Informational lecture	–	-	–
40	Calculate graph distances and metrics.	1	-	2	-	-	1	Seminar	-	-	-
41	Flat graph, bipartite graph, trees. The Problem of Vapor Combination	1	2	–	–	-	1	Informational lecture	–	-	–
42	Minimization of intersections of graph edges. Pairing problem	1	-	2	-	-	1	Seminar	-	-	-
43	Application of graph theory in AI systems. Spectral analysis of graphs. Basic numbers of graph theory. Graph coloring algorithm	1	2	–	–	-	1	Informational lecture	–	-	–
44	The basic numbers of graph theory (chromatic and cyclomatic numbers, connectivity components, skeleton number).	1	–	2	–	-	3	Seminar	Practical Work No3	Practical work	6
45	Determination of the maximum internally stable, minimum externally stable sets and cores of the graph	1	2	-	-	-	1	Informational lecture	-	-	-

No p/n	Topic of the lesson and issues under consideration	Semester	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
			Contact work				Independen t work				
			Lectures	Practical exercises	Laboratory Classes	Consulting					
46	Magu's method for finding maximum intrinsically stable, minimum externally stable sets and graph kernels	1	-	2	—	-	2	Seminar	—	-	—
47	The concept of a random graph, methods of generation	1	2	-	-	-	1	Informational lecture	-	-	-
48	Random graphs in AI systems	1	-	2	-	-	1	Seminar	-	-	-
49	Application of graph theory in AI systems. Graph neural networks (basics). Graph clustering concept	1	2	—	-	-	1	Informational lecture	-	-	-
50	Current trends in using the basics of graph theory in solving complex AI problems	1	-	2	-	-	1	Seminar	-	-	-
51	Rules and principles for inclusions-exclusions	1	2	-	-	-	1	Informational lecture	-	-	-
52	Permutations and combinations	1	-	2	-	-	1	Seminar	-	-	-
53	Basic rules and formulas of combinatorics	1	2	—	-	-	1	Informational lecture	-	-	-
54	Boundary control on graph theory and combinatorics	1	—	2	—	-	4	Seminar	Test No3	Test	14
Intermediate certification (exam)		1	—	—	—	-	36		Exam questions and tickets		40
Total Hours		1	54	54	-	-	108		—		

4.2. Plan of extracurricular independent work

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1. Fundamentals of Set Theory						
1	The Concept of Set	1	1	– study and repetition of the material of lectures;	1	[1]– [4], [12], [16]–[19]
2	Set Setting	1	1	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
3	Boolean Operations	1	1	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
4	Boolean functions	1	2	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
5	Set operations	1	2	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
6	Methods of mutual inclusion and from the opposite	1	2	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
7	Direct product of sets	1	3	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
8	Sets, equality of sets	1	3	\u2012 study and revision of lecture material; preparation for practical work No1;	3	[1]– [4], [12], [16]–[19]
9	Inversion and composition	1	3	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
10	Equality of Direct Products	1	4	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
11	Correspondences. Mapping operations	1	4	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
12	The image and prototype of the multitude	1	4	\u2012 study and revision of lecture material;	2	[1]– [4], [12], [16]–[19]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
13	Mapping properties	1	5	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
14	Features	1	5	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
15	Relationships. Relationship Operations	1	5	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
16	Relationship Equality	1	6	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
17	Basic Properties of Relationships	1	6	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
18	Basic Properties of Relationships	1	6	\u2012 study and revision of lecture material;	1	[1]– [4], [12], [16]–[19]
19	Fuzzy sets. Operations on fuzzy sets	1	7	\u2012 study and revision of lecture material;	1	[1]– [4], [16]–[19]
20	Inclusion and equality of fuzzy sets	1	7	\u2012 study and revision of lecture material;	2	[1]– [4], [16]–[19]
21	Fuzzy sets. Operations on fuzzy sets	1	7	\u2012 study and revision of lecture material;	1	[1]– [4], [16]–[19]
22	Boundary control on the basics of set theory	1	8	\u2012 study and revision of lecture material; preparation for the test No1;	4	[1]– [4], [12], [16]–[19]
Module 2. Algebra of Logic						
23	Elements of the Algebra of Logic	1	8	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
24	Work with basic logical identities and laws	1	8	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
25	Normal Forms of Boolean Functions	1	9	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
26	K-variable decomposition, SDNF, SCNF	1	9	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
27	Minimization of Boolean functions. Logic Circuits	1	9	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
28	Construction of Carnot maps of 2, 3, 4 variables. Logic Circuits	1	10	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
29	Minimizing Boolean Functions	1	10	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
30	Methods of Uncertain Odds, Quine McCluskey	1	10	\u2012 study and revision of lecture material; preparation for practical work No2;	3	[1]– [4], [16]–[19]
31	Minimization of polynomial SNF and in the Webb basis. The concept of a formal system.	1	11	\u2012 study and revision of lecture material	1	[1]– [4], [16]– [19], [21]
32	Minimization methods in polynomial SNF and in Webb's basis.	1	11	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
33	Post's theorem on functional completeness. 5 classes of Boolean functions.	1	11	\u2012 study and revision of lecture material	1	[1]– [4], [16]–[19]
34	Boundary Control in Set Theory and Algebra of Logic	1	12	\u2012 study and revision of lecture material; Preparation for the test No2	4	[1]– [4], [16]– [19], [21]
Module 3. Fundamentals of graph theory and combinatorics						
35	Basic concepts of graph theory and their interpretation. Fuzzy graph.	1	12	\u2012 study and revision of lecture material	1	[1], [5]–[13]
36	Types of graphs. Routes, chains, cycles, bridges, and hinges. Fuzzy graph.	1	12	\u2012 study and revision of lecture material	1	[1], [5]–[13]
37	Circumvention of the count. Special graph cycles (Euler and Hamiltonian). Graph cliques, isomorphism	1	13	\u2012 study and revision of lecture material	1	[1], [5]–[13]
38	Circumvention of the count. Construction of Euler and Hamiltonian chains and cycles. Graph cliques, isomorphism	1	13	\u2012 study and revision of lecture material	1	[1], [5]–[13]
39	Distance on the graph. Algorithms for finding the shortest paths. Graph metrics. The Pairing Problem	1	13	\u2012 study and revision of lecture material	1	[1], [5]–[13]
40	Calculate graph distances and metrics. The Pairing Problem	1	14	\u2012 study and revision of lecture material	1	[1], [5]–[13]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
41	Flat graph, bipartite graph, trees	1	14	\u2012 study and revision of lecture material	1	[1], [5]–[13]
42	Minimize graph edge intersections	1	14	\u2012 study and revision of lecture material	1	[1], [5]–[13]
43	Application of graph theory in AI systems. Spectral analysis of graphs. Basic numbers of graph theory. Graph coloring algorithm	1	15	\u2012 study and revision of lecture material	1	[1], [5]–[13], [15]
44	The basic numbers of graph theory (chromatic and cyclomatic numbers, connectivity components, skeleton number).	1	15	\u2012 study and revision of lecture material	1	[1], [5]–[13], [15]
45	Determination of the maximum internally stable, minimum externally stable sets and cores of the graph	1	15	\u2012 study and revision of lecture material	1	[1], [5]–[13]
46	Magu's method for finding maximum intrinsically stable, minimum externally stable sets and graph kernels	1	16	\u2012 study and revision of lecture material	2	[1], [5]–[13]
47	The concept of a random graph, methods of generation	1	16	\u2012 study and revision of lecture material	1	[1], [5]–[13]
48	Random graphs in AI systems	1	16	\u2012 study and revision of lecture material; preparation for practical work No3;	3	[1], [5]–[13]
49	Application of graph theory in AI systems. Graph neural networks (basics). Graph clustering concept	1	17	\u2012 study and revision of lecture material	1	[1], [5]–[14]
50	Current trends in using the basics of graph theory in solving complex AI problems	1	17	\u2012 study and revision of lecture material	1	[1], [5]–[14]
51	Rules and principles for inclusions-exclusions	1	17	\u2012 study and revision of lecture material	1	[4], [18]–[20]
52	Permutations and combinations	1	18	\u2012 study and revision of lecture material	1	[4], [18]–[20]
53	Basic rules and formulas of combinatorics	1	18	\u2012 study and revision of lecture material	1	[4], [18]–[20]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
54	Boundary control on graph theory and combinatorics	1	18	2012 study and revision of lecture material; Preparation for the test No3	4	[1], [4]–[15], [18]–[20]
Exam preparation					36	[1]–[21]
The total labor intensity of independent work in the discipline					108	–

V. EDUCATIONAL AND METHODOLOGICAL SUPPORT OF THE DISCIPLINE

The following forms of classes are provided for the discipline:

- informational lecture;
- Practical lesson

Along with traditional educational technologies, e-learning technologies and distance learning technologies can be used in the electronic information and educational environment of the Southern Federal University for the implementation of the discipline. Lectures and other forms of contact work between students and teachers can be conducted using Microsoft Teams, Cisco, Moodle (BigBlueButton), etc., which allows you to ensure online and offline interaction between the teacher and students within the discipline.

The main methods of current control are electronic accounting and control of students' educational achievements (using the tools of the point-rating system service; maintaining an electronic journal of progress, conducting electronic testing and using other means of control using the e-learning system).

5.1. References

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5.2. Further reading

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20. Zuev Yu.A. Modern Discrete Mathematics in Problems and Solutions. From enumerative combinatorics to cryptography of the XXI century. More than 700 problems with solutions. Moscow: Lenand, 2018. 304 p. (In Russian)
21. Glushan V. M. Mathematical Logic: Educational and Methodological Manual / V. M. Glushan ; Southern Federal University. – Rostov-on-Don : Taganrog, 2018. – 124 p.

5.3. Periodicals

- The scientific journal "Proceedings of SFedU. Technical Sciences" (http://www.izv-ti.tti.sfedu.ru/index.php/izv_tn);
- Scientific journal "Open Education" (<https://openedu.rea.ru/jour>);
- Electronic journal "IVTiO" (Informatics and Computer Engineering and Engineering Education) (<http://www.ivtio.sfedu.ru/>)

5.4. List of Internet resources

- Federal Portal "Russian Education" (<http://www.edu.ru/>);
- SFedU Electronic Resources Portal (<http://hub.sfedu.ru>);
- Electronic library system Lan (<http://e.lanbook.com>);
- Twirpx Information and Reference System (<https://www.twirpx.com/>);
- Electronic Library "University Library Online" (<http://biblioclub.ru>);
- Electronic Library (<http://www.rus-lib.ru>);
- NOU INTUIT (<http://www.intuit.ru>);
- Scientific Electronic Library (<http://elibrary.ru>).

VI. MATERIAL AND TECHNICAL SUPPORT OF THE DISCIPLINE

The following premises, equipment and software are used in the implementation of the discipline:

No p/n	Type of classroom lesson	Recommended audience	
		Type	Equipment
1	Lectures	lecture-type, seminar-type, current control, individual and group consultations, intermediate certification	– interactive whiteboard - 1 pc., teacher's computer - 1 pc.; – Microsoft Windows, Microsoft Office PowerPoint

No p/n	Type of classroom lesson	Recommended audience	
		Type	Equipment
2	Practical exercises	Laboratory of Information Technologies	– interactive whiteboard – 1 pc., teacher's computer – 1 pc.; – personal computers with Internet access – 10 pcs.; – Microsoft Windows, Microsoft Office, Microsoft PowerPoint
3	Independent work	Laboratory of Information Technologies	– interactive whiteboard – 1 pc., teacher's computer – 1 pc.; – personal computers with Internet access – 10 pcs.; – Microsoft Windows, Microsoft Office, Microsoft PowerPoint

VII. METHODOICAL INSTRUCTIONS FOR STUDENTS ON MASTERING THE DISCIPLINE

The discipline "Discrete Mathematics" is taught in the 1st semester. The educational process of teaching the discipline includes classroom classes (lectures and laboratory classes) and independent work. The final control in the discipline is an exam. Lecturers and teachers conducting training sessions control the attendance of all types of classroom classes. The system of university education is based on a rational combination of several types of classroom training (first of all, lectures and laboratory classes) and independent extracurricular and classroom work, each of which has a certain specificity.

If training sessions and/or intermediate certification are conducted using e-learning and DLT, then when organizing and conducting them, it is necessary to be guided by the Methodological Recommendations approved by the Order of the SFedU Rector No394 dated March 17, 2020, and the instructions in the section "E-Learning and Distance Educational Technologies" at <http://ictis.sfedu.ru/dot/>

Lectures and laboratory classes are carried out with the formulation of problematic issues that allow for discussions, which involves the active involvement of students in the educational process.

In the organization of the learning process in the discipline "Discrete Mathematics", both traditional, characteristic of the lecture-seminar form of education, and innovative (interactive, imitation, project) technologies are used. In the course of organizing training in the academic discipline, methods of problem-based learning, educational games, discussions, and testing methods are used.

Preparation for lectures. Before the first lecture, each student can download a PDF file of educational materials for each module of the discipline from the library, and then preliminarily work through the educational materials before each lecture. Each student is recommended to bring to the lecture any means for prompt viewing of these educational materials (laptop, tablet, smartphone, etc.) against the background of the lecturer's story electronic presentation in the form of slides on a large screen. These two sources of information eliminate the need to take lecture notes. But the student (if desired) can make important electronic or handwritten brief notes for himself

When working with lecture notes, it is necessary to take into account the factor that some lectures give answers to specific questions of the topic, others only reveal the relationships between phenomena, helping the student to understand the deep processes of the development of the subject under study, both in history and at the present time. Lecture note-taking is a complex type of university classroom work, which involves intensive mental activity of the student. A synopsis is useful when the most important things are written down and done by the student himself. Do not try to write down the entire lecture verbatim. Such "note-taking" does more harm than good. It is advisable to first understand the main idea presented by the lecturer, and then write it down. It is desirable to write on one page of the sheet or leaving margins on which later, when working independently with the synopsis, you can make additional notes, mark incomprehensible places. It is better to divide the lecture notes into points, observing the red

line. This will be greatly facilitated by the questions of the lecture plan proposed to teachers. It is necessary to pay attention to the accents, conclusions that the lecturer makes, noting the most important points in the lecture material with the remarks "important", "good to remember", etc. It is advisable to develop your own system of abbreviations, abbreviations and symbols. However, when working with the outline in the future, it is better to replace the symbols with ordinary words for quick visual perception of the text. When working on lecture notes, it is always necessary to use not only the textbook, but also the literature that is additionally recommended by the lecturer. It is such serious, painstaking work with lecture material that will allow you to deeply master the theoretical material.

The lecturer's teaching materials and presentations on the modules of the discipline have the following distinctive features: a system of abbreviations (abbreviations), highlighting important information (in bold, underlining, etc.). In addition, the educational material contains a glossary of the main terms of the subject area and control questions for the module, and the presentation uses accentuating highlighting in color and signs in the text and illustrations. Working with the mentioned materials before lectures, the student should use both the main and additional literature recommended by the lecturer. It is such serious, painstaking work with lecture materials that will allow you to deeply master the theoretical material of the discipline.

Preparation for practical classes. The main purpose of these classes is to form students' analytical, creative thinking by acquiring practical skills. Preparation for each practical lesson should begin with familiarization with the curriculum of the practical lesson, which reflects the content of the proposed topic. Careful thinking and study of the issues of the curriculum is based on the study of the current material of the lecture, and then the study of mandatory and additional literature. recommended for this topic.

A practical lesson is one of the main forms of organizing the educational process in technical specialties, aimed at the creative assimilation of the theoretical foundations of the academic discipline and obtaining practical research skills by setting, conducting, processing and presenting the results of an experiment on the basis of the practical use of VT means for observation, measurement, control, evaluation, and acquisition of skills for the experience of creative activity. Practical work is a specific educational task in the discipline being studied, performed in a practical lesson. The purpose of the practical lesson is the practical development of the content and methodology of the discipline studied by students using special means. The main objectives of the practical lesson are:

- \u2012 gaining experience in solving educational and research and real practical problems based on the studied theoretical material;

- \u2012 gaining experience in conducting an experiment; \u2012 mastering new methods of experimentation in the relevant branch of science, engineering and technology;

- \u2012 acquisition of skills and abilities to operate technical facilities and equipment;

- \u2012 formation of skills for processing the results of research;

- \u2012 analysis and discussion of the results obtained and formulation of conclusions;

- \u2012 development of the ability to logically comprehend independently acquired knowledge;

- \u2012 ensuring a rational combination of collective and individual forms of education.

Students who have confirmed their readiness to the extent of the requirements contained in the methodological instructions for practical work and (or) in the oral preliminary instructions of the teacher are allowed to perform practical work. If there are two completed, but not defended laboratory works on hand, the student is not allowed to perform the third one. During practical classes, students keep the necessary notes, draw up (at the request of the teacher) a final written report. In the first lesson of the cycle of practical work, the teacher should give specific instructions on the preparation and execution of reports in order to ensure uniformity. Depending on the characteristics of the cycle of practical classes, the report is compiled by each student individually, or a general report is drawn up by a subgroup of 2-3 students. At the end of practical work, students are required to submit a report to the teacher for verification with subsequent defense. By agreement with the teacher, it is allowed to submit a report on practical work for defense during the next practical lesson or within individual terms agreed with the teacher. It is allowed, in agreement with the teacher, to submit a report on practical work in electronic form. At the end of the practical lesson, the teacher evaluates the student's work by checking the report and defending it (interview).

The result of such work should be manifested in the student's ability to freely answer the theoretical questions of the practical workshop, participate in a collective discussion of the issues of the topic under study, and correctly perform the tasks proposed for implementation. In the process of preparing for practical classes, students need to pay special attention to the independent study of the recommended literature. With all the completeness of the lecture notes, it is impossible to present all the material in it due to the limit of classroom hours. Therefore, independent work with textbooks, manuals, scientific, reference literature, materials of periodicals and the Internet is the most effective method of obtaining additional knowledge, allows you to significantly intensify the process of mastering information, contributes to a deeper assimilation of the studied material, forms students' attitude to a specific problem.

Practical work. The student submits an individual assignment for verification by the teacher to assess the correctness of the completed tasks. If at least 60% of the maximum score is collected, the student defends his work by answering the teacher's questions, otherwise the work is returned to the student for revision. The main reference point for the defense of the task can be the interview questions given in the fund of assessment tools. Assessment of the individual task is carried out in accordance with the criteria specified in the fund of assessment tools. When drawing up an individual task, it is necessary to adhere to the following rules: 1) the report must contain a title page indicating the number of the option, an introduction; the main thematic sections (formulation of the goal of the work, formulation of the task according to the version of the individual task, description of the solution method(s), initial data and a detailed description of the procedure for solving the problem, conclusions, generalization of the results); conclusion (it should briefly formulate the results of the student's independent work, systematized conclusions on the main thematic sections, a reasonable assessment of the degree of achievement of the goal); a list of sources used (a numbered bibliographic list of literary sources and/or Internet resources drawn up in accordance with GOST); appendices (program listings and other additional material); 2) the report must comply with GOST in relation to bibliography, references, formulas, figures, tables, compliance with the rules of the Russian language, etc.; 3) the report is printed on a standard sheet of A4 paper. Margins of the printed sheet: left margin – 30 mm, right margin – 12-15 mm, top and bottom – 20 mm. Main font of the text – only Times New Roman, 14 font size, line spacing 1.5. Paragraph indent – 1.25 cm. The text of the work is set out on one side of the sheet. Justified text alignment. Each new section (1st level heading), with the exception of 2-4 level headings, starts with a new page; the same rule applies to other main structural parts of the work (introduction, conclusion, list of references, appendices, etc.); 4) when using materials borrowed from literary sources in the work, citing various authors, it is necessary to make appropriate references, and at the end of the work place a list of references. Not only quotations, but also an arbitrary presentation of the principal provisions borrowed from the literature are included with reference to the source. Quotations are marked with quotation marks. A reference to a literary source is made in the text in square brackets. It is a serial number of a literary source from the bibliographic list (it is possible to indicate the number of the page from which the quotation is taken). For example: [12, p.181]. A reference to several sources in a list in a row is performed as, for example, [10–15]. If the text of the work uses the ideas and thoughts of other authors, presented by them in different places of publication, then a reference to the source (sources) is made, and the page number is not indicated, for example: [7] or [2, 8-10]. Recommendations for working with literature. It is advisable to start working with literature with the study of general works on the topic, as well as textbooks and manuals. Next, it is recommended to proceed to the analysis of monographs and articles that consider certain aspects of the problems studied in the course, as well as official materials and unpublished documents (research papers, dissertations), which may contain the main issues of the problem under study. Work with sources should begin with introductory reading, i.e. review the text, highlighting its structural units. During introductory reading, bookmarks mark those pages that require more careful study. Depending on the results of the introductory reading, the further method of working with the source is chosen. If the solution of the problem requires the study of some fragments of the text, then the method of selective reading is used. Selected fragments or the entire text (if it is entirely related to the topic) require thoughtful, unhurried reading with a "mental study" of the material. Such reading presupposes the allocation of: 1) the main thing in the text; 2) the main arguments; 3) conclusions. Particular attention should be paid to whether the thesis follows from the arguments or not. It is also necessary to analyze which of the author's statements are problematic, hypothetical and

catch hidden questions. It is clear that the ability to work with the text in this way does not come immediately. The best way to learn to highlight the main thing in the text, to catch the problematic nature of the statements, to evaluate the author's position is comparative reading, during which the student gets acquainted with different opinions on the same issue, compares the weight and evidence of the arguments of the parties and concludes that this or that position is the most convincing. If there are different points of view on a particular issue in the literature, they cannot be rejected without understanding. If there are discrepancies between the authors, it is necessary to find a rational kernel in each of them, which will allow a deeper understanding of the subject of study and a more critical assessment of the issues under study. The next stage of working with literary sources is the creation of notes that record the main theses and arguments. You can make notes on separate sheets, which can then be easily systematized on individual topics of the course being studied. Another way is to keep thematic notebooks-notes on one topic. It is advisable to take notes of large special monographic works separately. Thus, when working with sources and literature, it is important to be able to: • compare, compare, classify, group, systematize information in accordance with a certain educational task; • generalize the information received, evaluate what has been listened to and read; - record the main content of messages; formulate, orally and in writing, the main idea of the message; • make a plan, formulate theses; • work in different modes (individually, in pairs, in a group), interacting with each other; • use abstract and reference materials; • control their actions and the actions of their friends, objectively evaluate their actions; • ask for help, additional explanations from the teacher, other students. • use linguistic or contextual guesses, dictionaries of various kinds, various kinds of hints, supports in the text (key words, text structure, preliminary information, etc.).

Preparation for intermediate certification. Students who have scored 38 or more points during the semester are considered to be admitted to the exam. Students who, for a good reason, could not score the required number of points in the current control of the module, can, in agreement with the teacher, eliminate debts before the start of the intermediate certification (exam). given in the fund of assessment tools. When studying the material related to a particular issue, you should carefully read the recommended literature, identify and consider various approaches to its solution, analyze their similarities and differences, possible advantages and disadvantages. When preparing for the exam, it is recommended to draw up a plan for answering the question and give examples of the use of the theoretical provisions under consideration in practice. It is especially important to show the examiner a confident command of the topic of each question, which is achieved by giving the necessary examples from related topics, attracting successful information from related topics, qualitatively assessing the state of the question, mentioning developments, details, names, chronology, etc.

VIII. FUND OF ASSESSMENT TOOLS

8.1. Passport of the Appraisal Tools Fund

Nop/ n	Competency Achievement Indicator	Name of the Valuation Tool
1	CPC-1.1 analyzes the natural science essence of problems arising in the course of professional activity	– practical works No 1-3 (execution, preparation of the report, defense of the report);
2	CPC-1.2 Solves standard professional problems using interdisciplinary and general scientific knowledge, methods of mathematical analysis and modeling	– Tests No 1-3 – exam questions and tickets.
3	PC-5.1 MF-5.1 Applies modern theoretical mathematics to develop new algorithms and formulate promising AI problems	– Test No 3 – practical works No 1-3 (execution, preparation of the report, defense of the report); – exam questions and tickets.

8.2. Test No 1

Type of assessment tool (subject of control): test work

List of controlled topics (modules): operations on sets, direct product of sets, correspondence, relation, fuzzy sets.

Form of study: full-time

An example of the formulation of tasks by levels of competence development:

№	Task Statement	Assessment of the task, points	
		Basic level	Advanced level
1	Prove the equality of sets by the method of mutual inclusion	3	3
2	Prove the equality of sets by the method of the opposite	3	3
3	Prove or disprove the equality of direct products of sets	4	4
4	Define mappings that have specific properties	2	0
5	Define relationships that have specific properties	2	0
6	Define fuzzy matches that have specific properties	0	2
7	Define fuzzy relationships that have specific properties	0	2
	Total:	14	14

Evaluation criteria for the test No1:

14-12 points: all the tasks of the practical part of the practical work are correctly completed, all control questions are answered correctly, a report on the work is provided in a timely manner.

9-11 points: all tasks of the practical part of practical work are correctly completed, answers to most of the control questions are given correctly, a report on the work is not provided in a timely manner, or in case of timely submission of a report, but with the presence of insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline.

7-8 points: not all, but more than 50% of the tasks of practical work are completed, some of the control questions are answered, there are insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline, a report on the performance of work is not provided in a timely manner.

0 points: less than 50% of the practical tasks of the practical part of the work have been completed, no answers to the control questions have been given, there are gross errors in the performance of practical tasks and/or answers to the control questions that contradict or distort the basic concepts of the discipline, a report on the completion of the work has not been provided.

8.3. Test No2

Type of assessment tool (subject of control): test work

List of controlled topics (modules): minimization of Boolean functions, logical circuits

Form of study: full-time

An example of the formulation of tasks by levels of competence development:

№	Task Statement	Assessment of the task, points	
		Basic level	Advanced level
1	Simplify a logical expression (reduce to a normal form of notation)	2	0
2	Implement a logical expression as a schema in a given basis	0	2
3	Minimize Boolean function using the Carnot card method	3	3
4	Minimize Boolean Function (Problem 3) by Quine-McClassky Method	5	5
5	Reduce to the pre-normal form of the formula	2	2
6	By the method of resolutions, to prove that the formula G is a logical consequence of the premises Fi	2	2
	Total:	14	14

Evaluation criteria for the test No2:

14-12 points: all the tasks of the practical part of the practical work are correctly completed, all control questions are answered correctly, a report on the work is provided in a timely manner.

9-11 points: all tasks of the practical part of practical work are correctly completed, answers to most of the control questions are given correctly, a report on the work is not provided in a timely manner, or in case of timely submission of a report, but with the presence of insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline.

7-8 points: not all, but more than 50% of the tasks of practical work are completed, some of the control questions are answered, there are insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline, a report on the performance of work is not provided in a timely manner.

0 points: less than 50% of the practical tasks of the practical part of the work have been completed, no answers to the control questions have been given, there are gross errors in the performance of practical tasks and/or answers to the control questions that contradict or distort the basic concepts of the discipline, a report on the completion of the work has not been provided.

8.4. Test No 3

Type of assessment tool (subject of control): test work

List of controlled topics (modules): basic numbers of graph theory, determination of maximum intrinsically stable, minimum externally stable sets and graph nuclei, graph spectral analysis, random graphs, rules and formulas of combinatorics

Form of study: full-time

An example of the formulation of tasks by levels of competence development:

№	Task Statement	Assessment of the task, points	
		Basic level	Advanced level
1	Draw a graph with a number of vertices ... and the number of ribs Set it in the set-theoretic form $G=(X,U,F)$.	1	0
2	Construct the skeleton of the graph (p.1) and specify it using the incident and contiguity matrices	2	0
3	Extract from the resulting skeleton (p.2) a subgraph on ... vertices, and an arbitrary sugraph containing ... ribs.	2	0
4	Draw a fuzzy graph with a number of vertices... and the number of ribs Set it in the set-theoretic form $G=(X,U,F)$.	0	1
5	Construct a graph skeleton (p.4) and define it using the incident and vertex contiguity matrices	0	2
6	Extract from the resulting skeleton (p.5) a subgraph on ... vertices, and an arbitrary sugraph containing ... ribs.	0	2
7	Prove that a complete graph ... is not planar	3	3
8	For a given graph, find by Magu the maximum internal and minimum externally stable sets, the numbers of internal and external stability, and the kernel.	6	6
	Total:	14	14

Evaluation criteria for the test No3:

14-12 points: all the tasks of the practical part of the practical work are correctly completed, all control questions are answered correctly, a report on the work is provided in a timely manner.

9-11 points: all tasks of the practical part of practical work are correctly completed, answers to most of the control questions are given correctly, a report on the work is not provided in a timely manner, or in case of timely submission of a report, but with the presence of insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline.

7-8 points: not all, but more than 50% of the tasks of practical work are completed, some of the control questions are answered, there are insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline, a report on the performance of work is not provided in a timely manner.

0 points: less than 50% of the practical tasks of the practical part of the work have been completed, no answers to the control questions have been given, there are gross errors in the performance of practical tasks and/or answers to the control questions that contradict or distort the basic concepts of the discipline, a report on the completion of the work has not been provided.

8.5. Practical Work No 1 (execution, preparation of the report, defense of the report)

Type of assessment tool (subject of control): defense of practical work

List of controlled topics (modules): calculus of sets, operations on sets; ordered sets; methods of construction and use in solving practical problems;

Form of study: full-time

Objective: to study and apply operations on sets, to construct proofs of identities with sets.

Tasks: construction of operations of intersection, union and difference of sets, solving practical problems.

Content: the construction of operations on sets (unification, intersection and differences), the construction of proofs of the identities of sets, using the method of two inclusions and the method of the opposite.

Criteria for evaluating practical work No1:

6-5 points: all the tasks of the practical part of the practical work are correctly completed, all control questions are answered correctly, a report on the work is provided in a timely manner.

4-3 points: all the tasks of the practical part of the practical work are correctly completed, answers to most of the control questions are given correctly, a report on the performance of work is not provided in a timely manner, or in the case of timely submission of a report, but with the presence of insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline.

2 points: not all, but more than 50% of the tasks of practical work are completed, some of the control questions are answered, there are insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline, a report on the performance of work is not provided in a timely manner.

0 points: less than 50% of the practical tasks of the practical part of the work have been completed, no answers to the control questions have been given, there are gross errors in the performance of practical tasks and/or answers to the control questions that contradict or distort the basic concepts of the discipline, a report on the completion of the work has not been provided.

8.6. Practical Work No 2 (execution, preparation of the report, defense of the report)

Type of assessment tool (subject of control): defense of practical work

List of controlled topics (modules): elements of the algebra of logic; normal forms of Boolean functions; minimization of Boolean functions; Carnot's cards;

Form of study: full-time

Objective: to acquire practical skills in constructing formulas for the logic of statements, as well as to analyze their properties, to know what DNF and KNF are, to be able to reduce the formulas of the algebra of logic to the SDNF and SCNF and minimize them using the laws of logic algebra, to acquire skills in solving Boolean equations, to gain practical skills in constructing Carno-Veitch maps of 2, 3 and 4 variables.

Task: to construct formulas for the logic of statements, with the subsequent analysis of their properties.

Content: to construct truth tables for a certain number of statements, to construct SNDF, SKDF, to find the minimum DNF. To give written answers to the control questions presented in practical work.

Criteria for evaluating practical work No2:

6-5 points: all the tasks of the practical part of the practical work are correctly completed, all control questions are answered correctly, a report on the work is provided in a timely manner.

4-3 points: all the tasks of the practical part of the practical work are correctly completed, answers to most of the control questions are given correctly, a report on the performance of work is not provided in a timely manner, or in the case of timely submission of a report, but with the presence of insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline.

2 points: not all, but more than 50% of the tasks of practical work are completed, some of the control questions are answered, there are insignificant errors in the performance of practical tasks and/or answers to control questions that do not contradict the basic concepts of the discipline, a report on the performance of work is not provided in a timely manner.

0 points: less than 50% of the practical tasks of the practical part of the work have been completed, no answers to the control questions have been given, there are gross errors in the performance of practical tasks and/or answers to the control questions that contradict or distort the basic concepts of the discipline, a report on the completion of the work has not been provided.

8.7. Practical Work No 3 (execution, preparation of the report, defense of the report)

Type of assessment tool (subject of control): defense of practical work

List of controlled topics (modules): introduction to graph theory; fuzzy columns; special graph cycles (Euler and Hamiltonian); trees; distance on the graph; algorithms for finding the shortest paths; graph metrics; minimization of graph edge intersections; graph coloring algorithm; Spectral analysis of graphs.

Form of study: full-time

Objective: to study the basics of set-theoretic and graphical representation of graphs, the simplest properties of graphs, to gain practical skill in defining and visualizing a graph on a plane, to consolidate the skills of building routes, circuits, cycles and hinges in graphs and the application of algorithms on graphs.

Objective: to consolidate knowledge of the basic concepts of graph theory, to acquire practical skills in visualizing graphs on a plane, as well as building routes, chains, cycles and hinges in graphs and applying algorithms on graphs.

Content: Application in practice of theoretical knowledge in the field of graph construction on a plane, as well as approaches to building routes of cycle chains and hinges in graphs. Determination of graph metrics for solving practical problems. Including the performance of case tasks 1-3 in the classroom.

Example of the wording of case task 1 by levels of competence development.

To ensure perimeter protection, choose a suitable mobile agent model with specified technical characteristics. A graph is created where each vertex represents an object in the system. Graph metrics are defined and compared to agent metrics. Thus, the most suitable agent model for perimeter protection is selected.

№	Task Statement	Assessment of the task, points	
		Basic level	Advanced level
1	Represent a given system as a graph model.	0,5	0
2	Represent a given system as a fuzzy graph model	0	0,5
3	For the resulting graph, form a distance matrix according to a given algorithm	1	1
4	Define graph metrics and perform source system analysis	0,5	0,5
	Total:	2	2

Example of the wording of case task 2 by levels of competence development.

For a multiprocessor system, determine the minimum number of registers required to store the initial, current, and final values of the program operands.

A graph is created where each vertex represents a variable or temporary value from the program. An edge connects two vertices if the corresponding variables are used at the same time and cannot be stored in the same register.

The compiler uses a coloring algorithm to assign each vertex (variable) a "color" corresponding to one of the available registers.

The basic condition is that no two adjacent vertices (variables used at the same time) can have the same color (register).

№	Task Statement	Assessment of the task, points	
		Basic level	Advanced level
1	Represent a given system as a graph model	0,5	0
2	Represent a given system as a fuzzy graph model	0	0,5
3	For the resulting graph, color the vertices, obtain independent subsets of vertices, and determine the chromatic number of the graph	1	1
4	Perform Source System Analysis	0,5	0,5
	Total:	2	2

Example of the wording of case task 3 by levels of competence development.

For a local area network, solve the problem of finding a minimum set of servers or routers from which the entire network can be managed.

A graph is created, where each vertex is a workstation. Using the Magu method, determine the minimally outwardly stable sets of vertices of the graph.

№	Task Statement	Assessment of the task, points	
		Basic level	Advanced level
1	Represent a given system as a graph model	0,5	0
2	Represent a given system as a fuzzy graph model	0	0,5
3	For the resulting graph, define externally stable sets of vertices	1	1
4	Perform Source System Analysis	0,5	0,5
	Total:	2	2

Evaluation criteria for case tasks:

2 points: the case task was completed correctly and completely, all control questions were answered, a report on the completion of work was provided in a timely manner.

1 point: the case task is not completed completely, but more than 50%, some of the control questions are answered, there are insignificant errors in the performance of the task and/or answers to the control questions that do not contradict the basic concepts of the discipline, a report on the performance of work is not provided in a timely manner.

0 points: less than 50% of the case task has been completed, no answers to the control questions have been given, there are gross errors in the performance of the task and/or answers to the control questions that contradict or distort the basic concepts of the discipline, a report on the completion of the work has not been provided.

8.8. Exam Questions and Tickets for Semester 1

Control period: 1 semester

Type of assessment tool (subject of control): exam

List of controlled topics (modules): Module 1. Fundamentals of Set Theory; Module 2. Algebra of Logic; Module 3. Fundamentals of graph theory and combinatorics.

List of exam questions

1. The concept of utterance. Logical formulas.
2. Quantifiers of community and existence. Their interconnection
3. The concept of a set.
4. Ways of setting sets.
5. Basic operations on sets.
6. A direct product of sets.
7. Set Design and Inversion
8. Composition of sets.
9. The concept of correspondence. The image and prototype of the multitude in a given correspondence.
10. Narrowing and continuation of correspondences
11. Basic properties of correspondences.
12. Function. Inverse function.
13. The concept of attitude.
14. Ways of setting relationships.
15. Operations on relationships.
16. Basic properties of relationships.
17. Equivalence relation.
18. Attitude of tolerance.
19. The concept of Boolean functions.
20. Logical schemes.
21. Logic of predicates.
22. Logical conclusion.
23. Non-classical logics.
24. Definition and methods of defining a graph.
25. Parts of the graph.
26. Properties of graphs.
27. Define a graph using matrices.
28. Graph connectivity.
29. A method for detecting all routes of a given length.
30. Graph metric.
31. The concept of an Euler graph. A necessary and sufficient condition.
32. The concept of a Hamiltonian graph. Sufficient condition theorem.
33. Internally stable set. The number of inner stability.
34. Graph coloring page.
35. Isomorphism of graphs.
36. Match combinations in columns.
37. Outwardly stable set. The number of external stability.
38. Spectral analysis of the graph.
39. Random graphs.
40. Fuzzy set, fuzzy set belonging function.
41. Unclear statement. Simple and composite fuzzy statements.
42. The degree of equality and the degree of inclusion of two fuzzy sets. Fuzzy equal fuzzy sets.
43. The concept of r-sampling. R-permutations with and without repetitions.
44. The concept of R-sampling. R-combinations with and without repetitions.
45. Placement and filling.

Sample exam ticket

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION
Federal State Autonomous Educational
Institution of higher education
"SOUTHERN FEDERAL UNIVERSITY"

EXAM TICKET No _____

In the discipline "**Discrete Mathematics**"

Structural unit: Institute of Computer Technologies and Information Security

Area of study: 10.03.01 Information Security

1 Question

2 Question

3 Question

Compiled by _____ A.V. Bozhenyuk, L.A. Gladkov, E.R. Zyablova

Chairman of the UMC ICTIB _____ A.E. Lyz
(signature)

"__" _____ 2025